Regulation, efficiency, 3-phase transformers

Practice questions · Day 05 · Wed 28 Oct · pack M5

5. Practice

From your course files

Jun24 Q2c (past exam, 8 marks). Three-Phase Transformers. A three-phase transformer is rated at 16 kV/220 kV, 150 MVA. It has the following equivalent circuit parameters (referred to the primary winding where appropriate): primary resistance $R_1 = 0.08$ ohms, secondary resistance $R_2' = 0.06$ ohms, primary leakage reactance $X_1 = 0.14$ ohms, secondary leakage reactance $X_2' = 0.14$ ohms, magnetising reactance $X_M = 14$ ohms, core-loss resistance $R_C = 250$ ohms.

i. Draw a simplified equivalent circuit for the transformer showing the values for each of the equivalent circuit parameters as impedances. [2 marks]
ii. Consider the transformer in the open-circuit condition with rated input voltage. Find the input current magnitude and (real) power. [4 marks]
iii. Consider the open-circuit condition as discussed above, if the input voltage was halved, by what factor (e.g. x0.25, x0.5, x1, x2 or x4) would you expect the following quantities to change by? Input current magnitude; input real power; input reactive power; input power-factor. [2 marks]

2023 quiz Q5 (quiz, 5 marks). Transformer Losses. Consider a three-phase transformer with the following simplified single-phase equivalent circuit parameters referred to the primary: magnetising reactance X_M: 9,100 ohms; core loss resistance R_C: 100,000 ohms; total winding resistance R: 8.5 ohms; total leakage reactance X: 50 ohms. Under a particular operating condition it is known that the: primary voltage is 11.7 kV; secondary current referred to the primary is 6.8 A; input power is 115 kW. Using the simplified equivalent circuit, calculate the copper losses, the iron losses and hence the transformer efficiency.

2023 quiz Q6 (quiz, 9 marks). Transformer Analysis. A three-phase 50 kVA, 3.3 kV/415 V transformer has the following simplified single-phase equivalent circuit parameters referred to the primary: total leakage reactance 35 ohms, total winding resistance 4 ohms, magnetising reactance 2,300 ohms, core loss resistance 18,000 ohms. The transformer is delivering 27 kW at a power-factor of 0.78 lagging at rated output voltage.

a. Calculate the magnitude and phase of the secondary phase voltage V_2' and current I_2' referred to the primary. [4 marks]
b. Under a different load condition, it is known that the secondary phase voltage $V_2' = 2100\angle 0^\circ$ V and that the secondary phase current $I_2' = 7.7\angle -25^\circ$ A (both referred to the primary). Draw a simplified equivalent circuit for the transformer referred to the primary and hence calculate the required primary (line) voltage magnitude. [5 marks]

2023 quiz Q7 (quiz, 4 marks). Short-Answer Questions. Answer the following question using words only (that is, without using sketches or equations). In designing a transformer using a given magnetic core, what is the process for selecting the number of turns on the primary winding for a given primary voltage? Carefully explain the results of using too few turns. How about if too many turns are used?

PQ 3.29 (practice questions). The following tests are performed on a single-phase 240V:24V, 2400VA transformer. Short-circuit test. Secondary winding is short-circuited. Primary measurements are 27 V, 9.5 A, and 55 W. Open-circuit test. Primary winding voltage is 240 V, current is 1.2 A, power is 45 W.

a. Find the simplified equivalent circuit parameters.
b. The transformer is delivering rated output at a power-factor of 0.87 lagging at rated output voltage. Calculate the secondary voltage and current referred to the primary.
c. Find the required primary voltage corresponding to the case in part (b).
d. Find the output power, copper loss, iron loss and efficiency of the transformer corresponding to the case in part (b).

PQ 3.30 (practice questions). A three-phase 3000 VA, 415V/48V transformer has the following simplified single-phase equivalent circuit parameters referred to the primary: total leakage reactance X 8.5 ohms, total winding resistance R 0.95 ohms, magnetising reactance X_M 510 ohms and core loss resistance R_C 3300 ohms. The transformer is delivering rated output at unity power-factor at rated output voltage.

a. Calculate the secondary phase voltage and current referred to the primary.
b. Calculate the required supply voltage magnitude.
c. Calculate both the total copper winding loss and total core loss.
d. Hence calculate the transformer efficiency.

PQ 3.31 (practice questions). The following tests are performed on a three-phase 15kV:135kV, 100 MVA star-connected transformer. Short-circuit test with secondary winding short-circuited. Primary measurements are 95 V, 500 A, and 12 kW. In the open-circuit test the primary winding voltage is 415 V, current is 15 A, power is 800 W.

a. Find the simplified equivalent circuit parameters referred to the primary winding.
b. Assuming the transformer is delivering 50 MW at a power-factor of 0.8 lagging at rated output voltage, calculate the secondary phase voltage and current referred to the primary and hence the required input voltage.
c. The transformer efficiency corresponding to the operating case described in part (b).

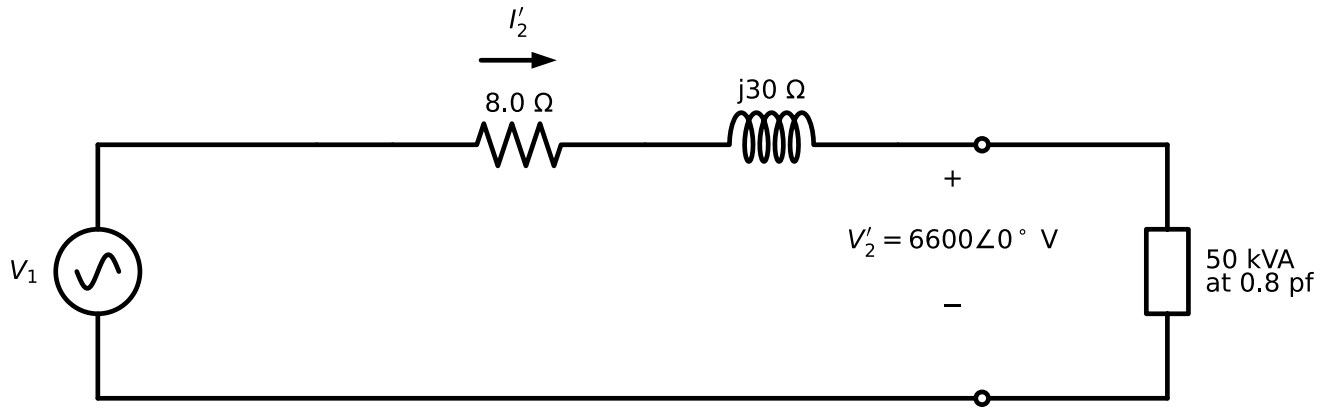
Jun24 Q4c(ii) (past exam, 3 marks). Answer the following questions using words only (that is, without using any sketches, drawings or equations).

ii. Consider a transformer operating from an AC supply. Carefully explain what are the main types of losses in the transformer, how are they modelled in the simplified equivalent circuit, and how do these vary with output power. [3 marks]

New questions

N1. Loss scaling and maximum efficiency [5 marks]. A single-phase 10 kVA, 2400/240 V transformer has an iron loss of 120 W at rated voltage and a full-load copper loss of 180 W. Assume rated output voltage throughout.

a. Find the efficiency at full load, 0.8 pf lagging. [2]
b. Find the efficiency at half load, 0.8 pf lagging. [1]
c. Find the load (kVA) at which efficiency is maximum, and the maximum efficiency at unity pf. [2]

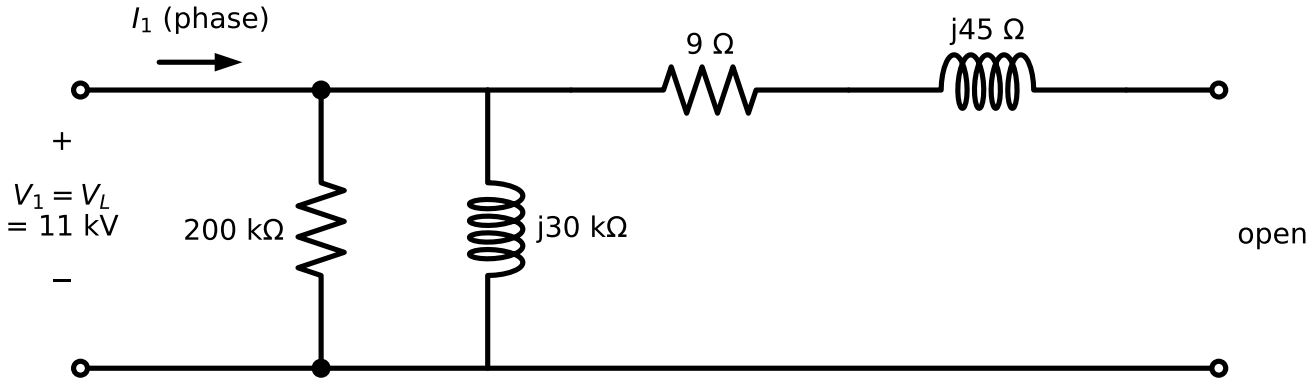


referred to the 6600 V primary; shunt branch neglected

Circuit for New question 2: series 8.0 Ω + j30 Ω (shunt branch neglected) delivering rated kVA at V_2′ = 6600∠0° V

N2. Regulation with lagging and leading loads [6 marks]. A single-phase 50 kVA, 6600/415 V transformer has $R = 8.0\ \Omega$ and $X = 30\ \Omega$ referred to the primary. The shunt branch can be neglected for the voltage calculation. The transformer delivers rated kVA at rated output voltage.

a. At 0.8 pf lagging, find the required V_1 and the regulation. [3]
b. Repeat for 0.8 pf leading. Explain the sign of the answer in one sentence. [2]
c. Using the approximate formula, find the leading pf at which the regulation is zero. [1]



one delta phase, referred to the primary

Circuit for New question 3: one delta phase on open circuit, 200 kΩ ‖ j30 kΩ across 11 kV, series 9 Ω + j45 Ω carrying no current

N3. Delta-connected primary on open circuit [6 marks]. A three-phase 500 kVA, 11 kV/415 V transformer has a **delta**-connected primary. The per-phase parameters, referred to the primary delta phase, are $R = 9\ \Omega$, $X = 45\ \Omega$, $R_C = 200\ \text{k}\Omega$ and $X_M = 30\ \text{k}\Omega$.

a. With rated voltage and the secondary open, find the phase current, the line current, the total input power and the pf. [3]
b. Express the OC line current as a % of rated line current. [1]
c. The supply falls to 5.5 kV. By what factors do the line current, P, Q and pf change? Would you expect the real transformer to follow these factors exactly? [2]

N4. Full chained analysis from test data [10 marks]. A three-phase, star-connected 1 MVA, 11 kV/433 V transformer gives these primary test results:

- OC: 11 kV, 3.0 A, 2.4 kW.
- SC: 550 V, 52.5 A, 9.0 kW.

a. Find R, X, R_C and X_M per phase. [3]
b. It delivers rated kVA at 0.85 pf lagging at rated output voltage. Find V_2', I_2', the required primary line voltage and the regulation. [3]
c. Find the efficiency for case (b). [2]
d. Find the kVA at which efficiency is maximum (at 0.85 pf) and the value of η_{max}. [2]

N5. Short answer: no-load current and the Prac 2 evidence [4 marks]. In words only:

a. Why is the input current of a transformer not zero when its secondary is open-circuited? [2]
b. In Prac 2 the open-circuit current rose from 0.04 A at 100 V to 0.66 A at 240 V. Explain why it rose so much more than the voltage. [2]

6. Quick recall check

1. A transformer runs at half its rated load. By what factor does the copper loss change, and by what factor does the iron loss change?
2. In the simplified circuit, which voltage do you use for the iron loss, and which current for the copper loss?
3. State the condition for maximum efficiency and the formula for the per-unit load at which it occurs.
4. On open circuit, the supply voltage is halved. Give the factors for current, P, Q and pf.
5. A star-connected transformer has a 33 kV line rating. What V_2' do you use for "rated output voltage", and how do you get the required supply line voltage from V_1?